

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with some lines of code in green and blue. At the bottom, a Windows taskbar is visible with various application icons like Chrome, Photoshop, and others. The text is overlaid on this background.

Introduction To Programming Using C++ Level 2

Library Create Your Own Library



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Library Create Your Own Library

Definition

- ◆ A Library is a collection of reusable Functions and code organized in one or more files.
- ◆ A Library allows us to reuse Functions instead of writing the same code again.
- ◆ A Local Library is a library created inside our project and used within that project.

Explanation

- In C++, we can create our own Header File to store reusable Functions.
- Header Files usually have the `.h` extension.
- We can organize related Functions inside a Namespace.
- The Namespace helps group related Functions and prevents naming conflicts.
- The Header File can then be included in another source file using `#include`.

Procedures

1. Open the Solution Explorer.
2. Go to Header Files.
3. Right-click Header Files.
4. Select Add → New Item → Header File (.h).
5. Give the Header File a suitable name, such as `Arrays.h`.
6. Visual Studio creates the Header File with `#pragma once`.
7. Create the required Functions inside a Namespace.
8. Include the Header File in the source file using `#include`.
9. Call the Functions using the Namespace name.

💡 Example

◆ Create a Header File called: `Arrays.h`

◆ Add the following code:

```
#pragma once
namespace Arr
{
    void ReadArr(int arr[], int length)
    {
        cout << "Enter the elements:\n";
        for (int i = 0; i < length; i++)
            cin >> arr[i];
    }
    void PrintArr(int arr[], int length)
    {
        for (int i = 0; i < length; i++)
            cout << arr[i] << ' ';
    }
}
```

◆ Then include the Library in the source file:

```
#include "Arrays.h"
```

◆ We can then call the Functions using the Namespace:

```
Arr::ReadArr(arr, 6);
```

```
Arr::PrintArr(arr, 6);
```

Note

- ◆ `#pragma once` helps prevent the Header File from being included multiple times in the same compilation unit.
- ◆ The `::` operator is called the Scope Resolution Operator and is used here to access Functions inside the `Arr` Namespace.

⊘ Warning

- ◆ A Local Library created inside a project is not automatically available to other projects.
- ◆ If another project needs the same Functions, the Library or its source code must be made available to that project.



Comparison

▨ Using a Library

- ✓ Functions can be written once and reused.
- ✓ Related Functions can be organized together.
- ✓ Makes the code easier to maintain and reuse.

▤ Writing Everything in the Main File

- ✓ Simple for very small programs.
- ✓ Can become difficult to manage as the project grows.
- ✓ Repeated code may need to be written multiple times.

✦✦ Summary

- ✓ A Library is a collection of reusable code and Functions.
- ✓ We can create our own Header File such as ``Arrays.h``.
- ✓ A Namespace can be used to organize the Functions.
- ✓ ``#include`` allows us to use the Header File in another source file.

Thank you 😊